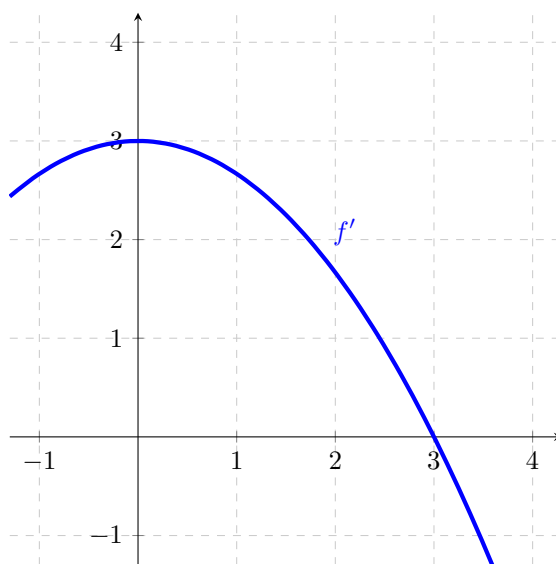


Problem 4

Problem 4

Problem 1 Consider the graph of $f'(x)$ given below. Suppose you know that $f(3) = 7$. Can you approximate $f(2.98)$ and $f(3.02)$? Explain your answer. Are these overestimates or underestimates?



Explanation. $f(3) = 7$ and $f'(3) = 0$

$$L(x) = 7 + 0(x - 3) = 7$$

This is a constant function, and so our approximations are $f(2.98) \approx 7$ and $f(3.02) \approx 7$. These are overestimates for the graph of f is concave DOWN on the intervals $(2.98, 3)$ and $(3, 3.02)$.